Empowering Development through Data-driven Innovation and Information Technology Solutions

Base IMIS Field Data Collection User Manual - Enumerator

Integrated Municipal Information System (IMIS)

Innovative Solution Pvt. Ltd (ISPL)

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Chapter 1 INTRODUCTION

This is an operational training manual for the survey enumerators. An enumerator is the person who will collect the field data using the developed mobile app. The prime responsibility is to gather field data ensuring optimal data quality and communicate with the assigned supervisors in case of any issues arise during the field survey. As an enumerator, the person must be aware of the purpose of the survey and its importance, interact with the respondent with friendliness, listen and record respondent's answers on the mobile device patiently. Furthermore, enumerators must ensure the highest ethical standards by treating all information that is gathered/observed in the field with the highest confidentiality.

The manual will give a brief idea about what the survey enumerators should do, and how to use the Field Survey mobile app in order to collect survey data. The key tasks of the enumerator are:

a. **Add Building Details:** The enumerator needs to add the details of the buildings that have been digitized previously through the survey application. (ref. to 2.1)

b. **Synchronization of Data**: The enumerator must periodically sync their data through the cloud. (ref. to 2.2)

1.1. Field Survey Application

The Field Survey Application has been developed using an open-source ecosystem comprising QGIS, Mergin and Input. The chosen ecosystem has numerous advantages that are essential for geo-spatial data collection e.g., ability to display spatial data, ability to navigate to spatial features using the device GPS, ability to capture geographic features, data storage local storage as well as cloud service, versioning control as well as syncing through cloud, and so on.

Forms and data preparation were done in open source QGIS Software and synchronized to the Input using the Mergin plugin and repository. Mergin is a collaborative cloud platform that lets users store and synchronize projects across multiple mobile and desktop clients. Enumerators will use Input for surveys to capture data in the field. The field coordinator will be provided access to share the mobile application with other enumerators via Mergin.

Input is specifically designed to capture geo-information easily through a mobile/tablet. It is a simple survey app that allows users to capture geo-spatial and attribute data on the field, and is available on both iOS & Android (Windows, Linux and MacOS is available). The application is based on QGIS; so, many functions that a user is able to do in QGIS can be done in Input as well.

1.2. Workflow

The workflow of the mobile application is shown below. Initially, a project is created in QGIS where the desired forms/questionnaires are created with various map themes, etc. The project is then uploaded to Mergin. The Field Coordinator of the project can share the project to different enumerators via Mergin. The enumerators can download the project onto their mobile phones and use it to collect data which can be synchronized to the main project easily.

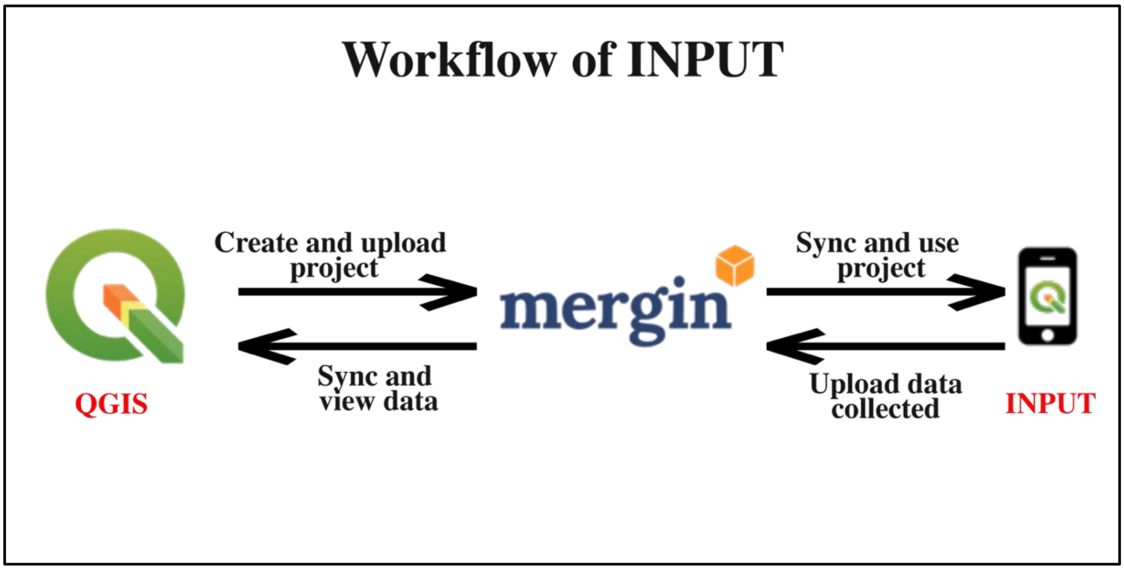


Figure 1 Input Workflow

Chapter 2 MOBILE APPLICATION FOR SURVEY

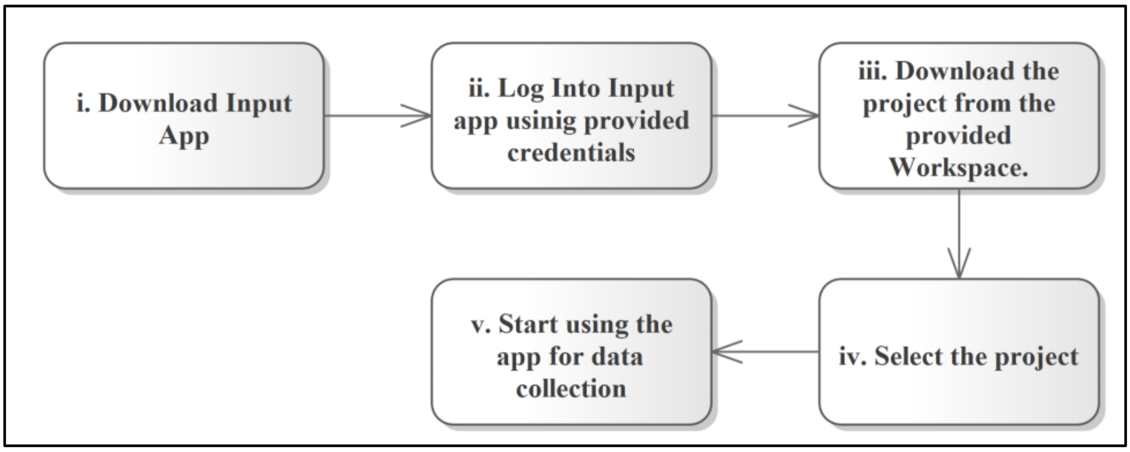


Figure 2. How to use Input

The flow chart above gives a general idea on how to use the Input mobile application for field survey A detailed step-by-step procedure is described below.

i. Download Input app

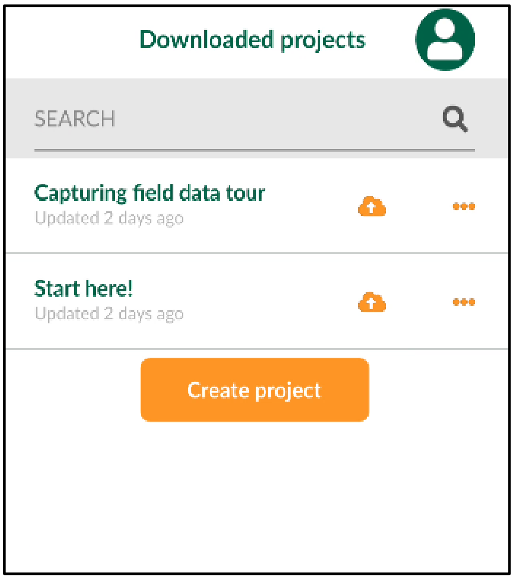
Input mobile app is available in both Android and iOS devices, however the android version is more stable in comparison to iOS thus iOS version is preferable.

- Go to the playstore/ app store
- Search for Mergin Maps
- Download the app

Figure 4 Download Input App

ii. Log into Input using provided credentials

- Locate the app on your device and open the app.
- Select on the User Profile icon on the top right of the app.
- Enter the username and password provided



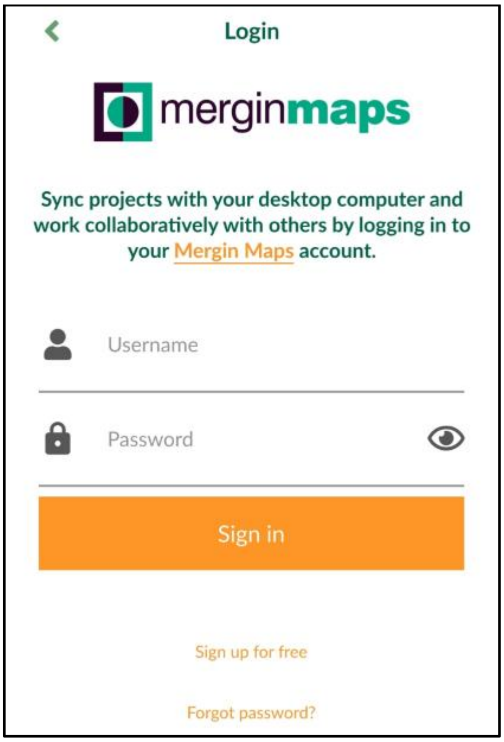


Figure 5 Input App Mergin Login

Note: The menu bar consists of 3 options:

a. Home: Consist of all the downloaded projects in the device.
b. My project: Projects of the user and hosted in Mergin account.
c. Explore: List of public projects on the Mergin service.

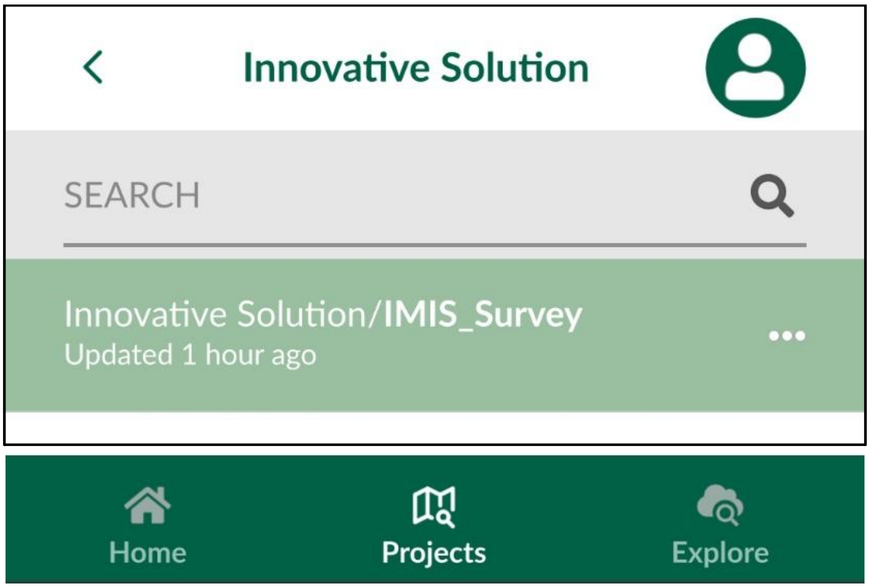
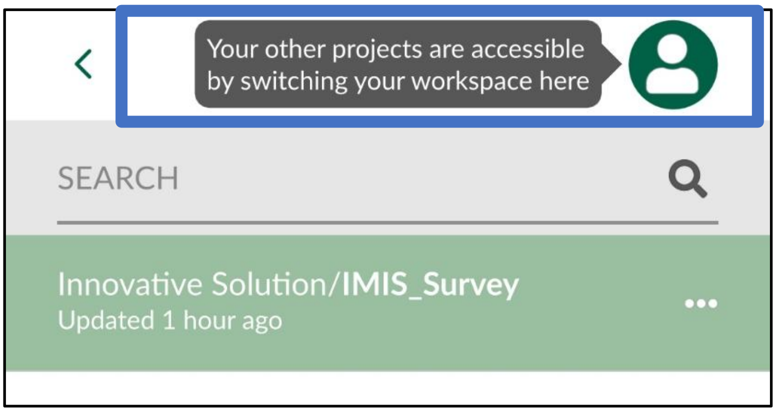


Figure 6 Input App Interface

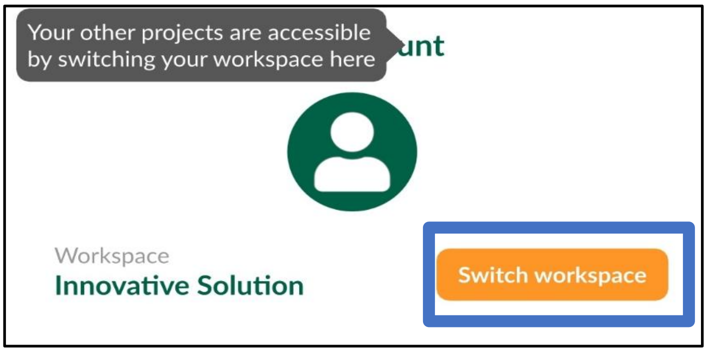
iii. Download the project from Shared with me

The Field coordinator will give access to the field project through Mergin based on their usernames. The shared project needs can be assessed through the workspace.

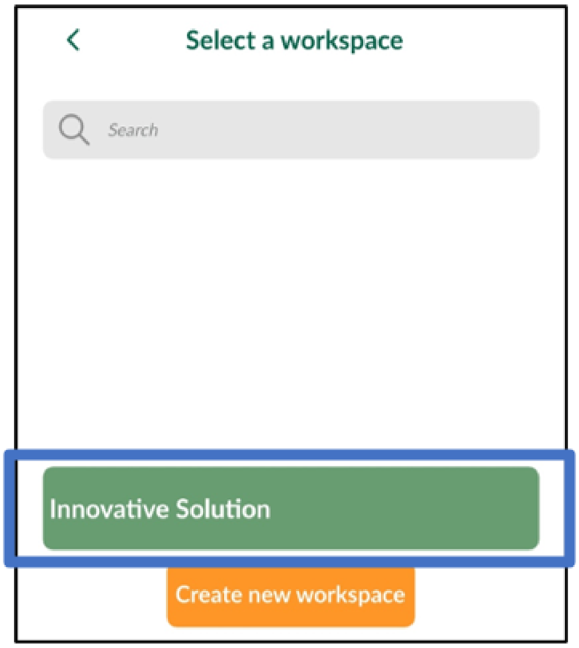
- Select on the My Account Option



- Select on the Switch Workspace option.



- Select the desired shared workspace.



- Click on the download button right next to the project name and the project will be downloaded to the local device.
- The user must check the project periodically for any changes to the project and sync with the main project by clicking on the sync button.

iv. Select the project.

After downloading the project to the local device, the project will appear on the home menu.

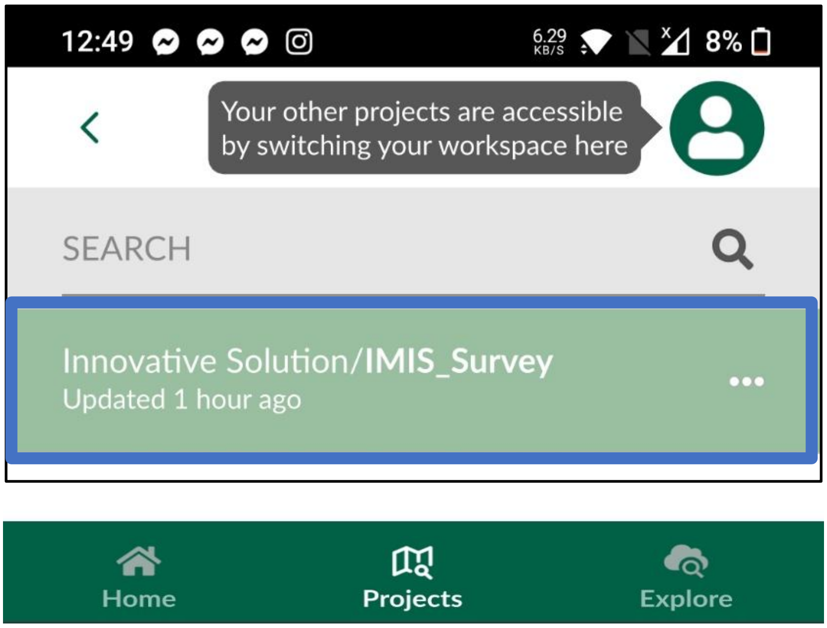


Figure 7 Home Tab

v. Starting using the app for data collection

When you start or open a project you are welcomed by a map with your location centered with already digitized buildings.

2.1. Adding Buildings Details

The buildings have been digitized and stored in the survey application and each building has already been assigned a BIN no. The enumerator should collect the data for the digitized buildings through the survey applications. To collect the data, follow the step below:

i. The buildings that are yet to be filled are in red color. Once the enumerator fills all the data on the building, the color changes from red to green. It is to be noted that the enumerators should not delete any features displayed in the map.

Figure 8 Field Survey App Interface

ii. Click on the building. This will prompt a dialogue box with BIN No. as the label. Click on the Edit Icon to open the forms.

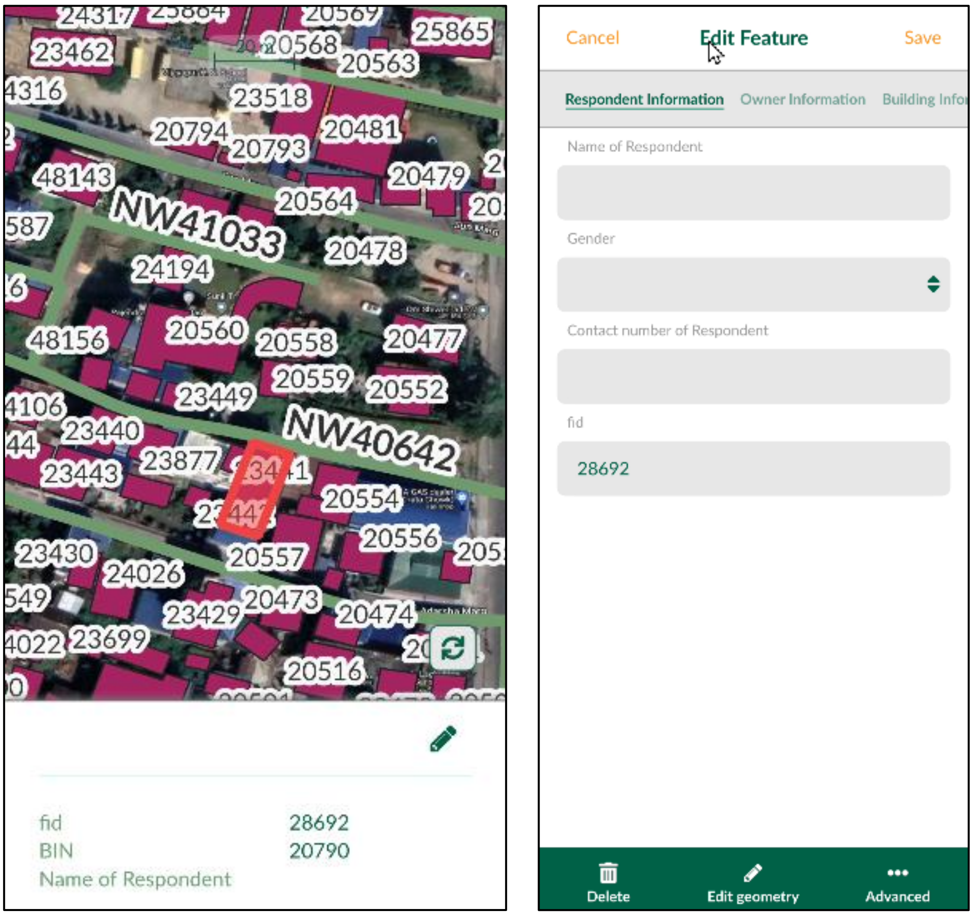


Figure 9 Selecting the Building

iii. The questions that are compulsory to be filled are in red color while those that are not compulsory are in black color. There are various skip logics built into the form, such that different form fields are hidden/ shown based on the input chosen. E.g., If the option *Others* is selected in any of the present drop downs, a new form field *Please Specify* is prompted, there are different form fields prompted according to the *presence of toilet*, etc.

Note: For detailed explanation of the form fields and skip logic, please refer to the questionnaire attached with this manual.

iv. There are six tabs mainly Respondent Information, Owner Information, Building Information, Toilet and Containment Information, SWM Information and Water Supply Information. The enumerator must fill the information in the aforementioned order by selection of the appropriate tabs and ensure the collected information is accurate before moving on to the next tab.

a. Respondent & Owner Information

The Respondent & Owner Information consists of the basic information of the respondent such as Respondent and Owner such as the Name, Gender, and Contact.

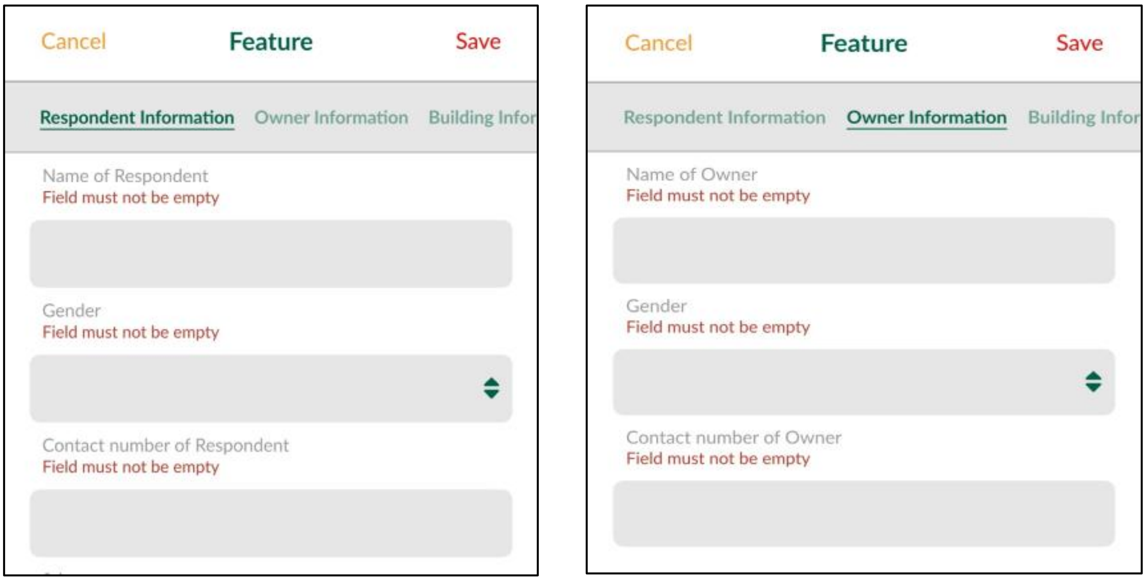


Figure 10 Respondent and Owner Info

b. Building Info

The Building Info consists of detailed information related to the building such as ward, location name, road ID, road name and so on. Note: the road ID and name can be identified through the map interface. Upon selecting the road, the corresponding road ID and name is displayed.

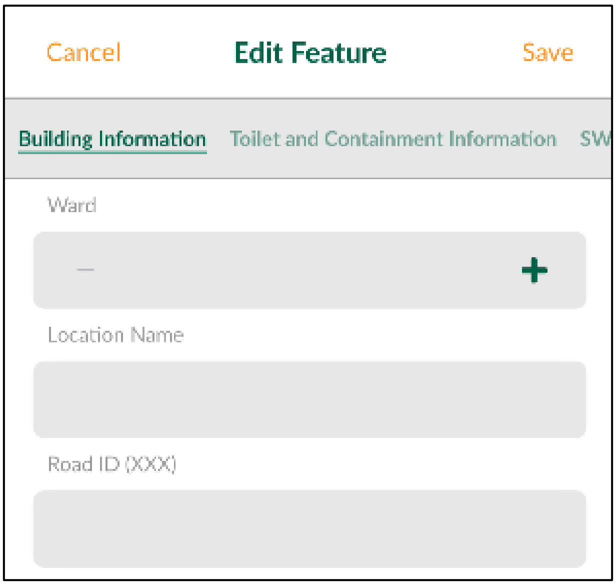


Figure 11 Building Info

c. Toilet and Containment Information

The Toilet and Containment Information consists of the basic information that is to be collected with respect to the status of the toilets. If toilets are absent, the information related to the defecation areas are prompted, and if they are present, information related to containment and other details are prompted.

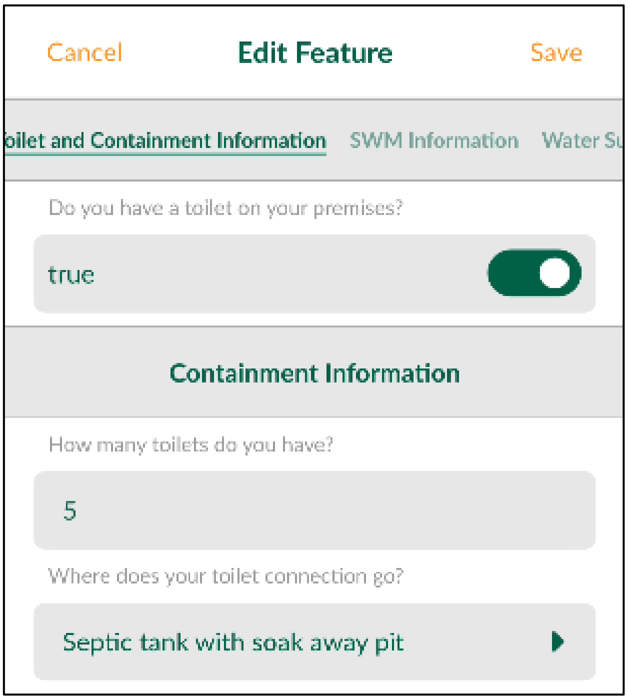


Figure 12 Toilet & Containment Information

d. SWM Info

The SWM Info consists of the basic information that is to be collected with respect to Solid Waste Management.

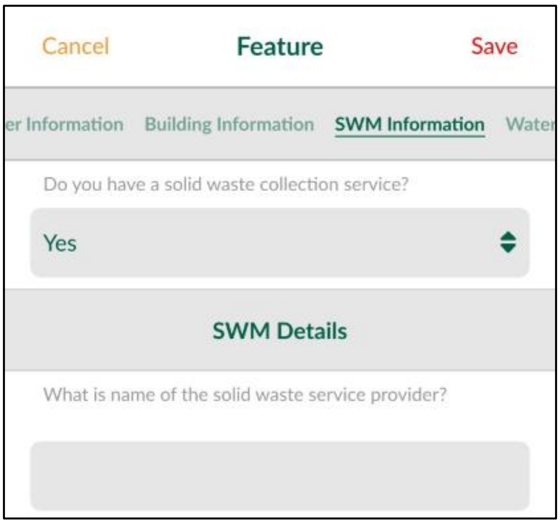


Figure 13 SWM Info

e. Water Supply Info

The Water Supply Info consists of the basic information that is to be collected with respect to the Water Supply of the building. If the Municipal/ Public water supply option is selected, then further form fields are prompted.

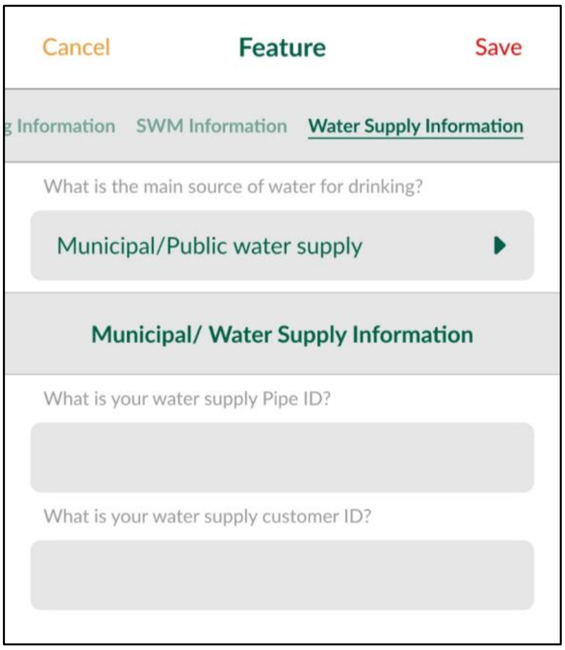


Figure 14 Water Supply Info

Once the form is filled select the save button to store the information. Once the button is pressed, the information is stored on the local device only. To synchronize, refer to no.4. The color of the digitized buildings turns green, this is done to help the enumerators identify which buildings have been filled and which have not.

2.2. Synchronization of Data

Once the enumerator has entered their data, they have to send the data to the cloud or to the database periodically to ensure the data is synchronized with the cloud database. To do that go to the homepage of the Input app and click the sync button right next to the name of the project. This can also be done through the Map Interface as well. The sync button can be found on the bottom right of the screen.

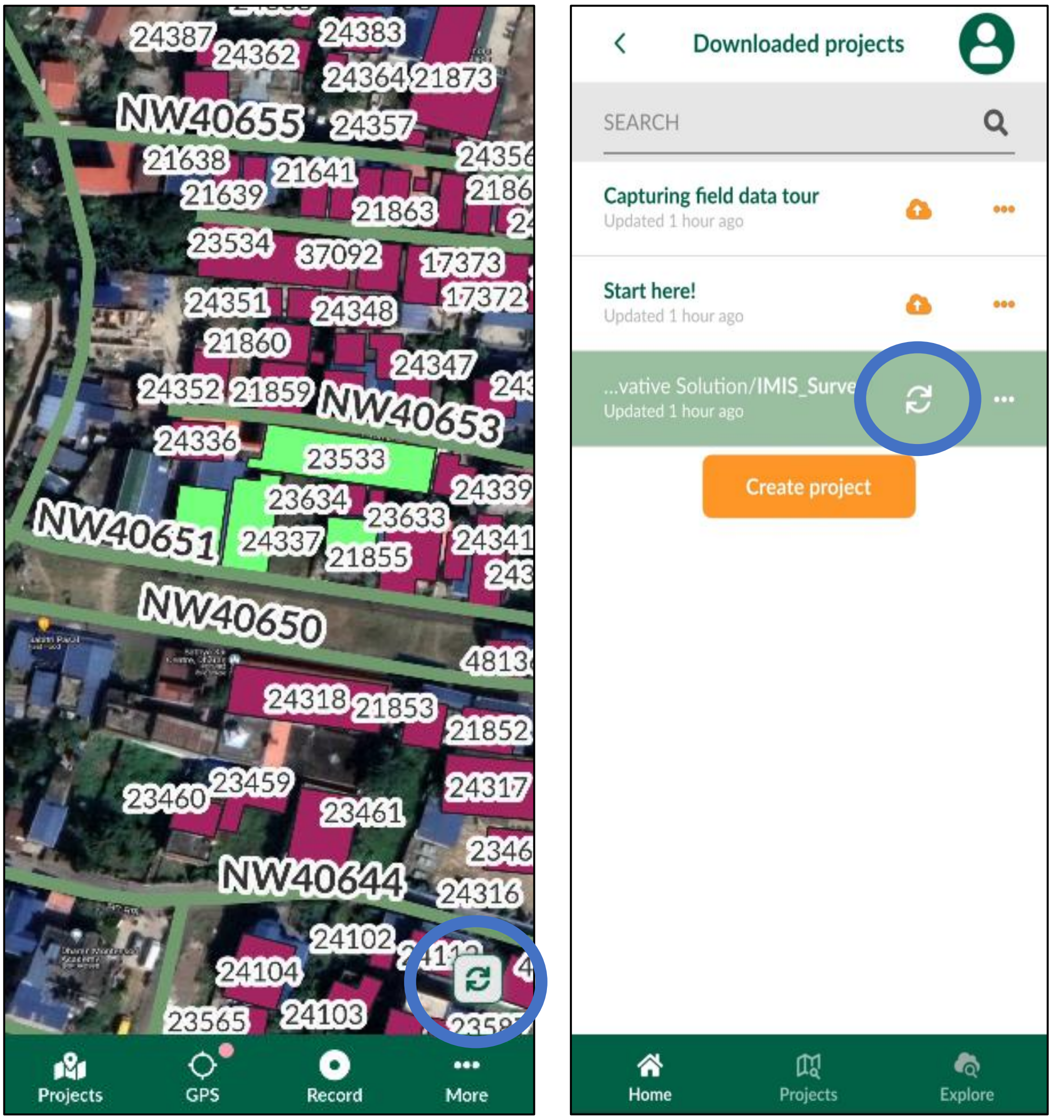


Figure 15 Synchronization Options